



Combining ENMAP Hyperspectral Imaging and Machine Learning for Land Use/Cover Classification

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Abstract

Imaging spectroscopy enables detailed recording of the Earth's surface and ecosystems' properties at a global level. Due to the specifications of the hyperspectral sensors, it becomes possible to utilize the hyperspectral data in a multitude of applications, among which the mapping, monitoring and understanding of land use and land cover (LULC). The Environmental Mapping and Analysis Program (EnMAP) is a German satellite mission that aims to monitor and characterize the Earth's surface on a global scale providing high-quality data with the advantage of open access. This study explores the capabilities of the EnMAP satellite in LULC mapping. The goal of the study is to leverage the advantages and potential of state-of-the-art machine learning algorithms, in particular of Support Vector Machine (SVM) and Random Forest (RF), in combination with EnMAP's hyperspectral data, for mapping LULC. The study area is characterized as a typical Mediterranean environment. The thematic LULC maps accuracy was assessed using statistical methods and further analysis was conducted through McNemar's statistical test to evaluate the statistical significance of the differences in the results. All in all, the SVM algorithm proved to be more accurate than RF algorithm, with an overall accuracy of 91% compared to 86%, respectively. The McNemar's test confirmed the higher accuracy of the SVM algorithm's results, at least this was the case in our study. The study's findings highlighted EnMAP's hyperspectral data promising role of in the field of LULC classification. This study adds valuable information towards evaluating EnMAP satellite data in producing thematic maps in typical Mediterranean environments and allows for comparisons with similar data sets in other areas or with data from other sensors in similar environments.

Keywords EnMAP · Hyperspectral remote sensing · Machine learning · Land use/land cover mapping

1 Introduction

The Earth's surface and the ecosystems on it are being disturbed by various factors related to both natural causes and human activity [1]. The rapid population growth combined

with economic growth have exponentially increased the need to exploit natural resources and thus the rate of land use/cover change [2, 3]. Accurate LULC maps allow better assessment of risks and design of mitigation measures [4]. Quantifying changes in LULC is an important means to understand land transformation and its management [5, 6].

Remote sensing is a field that has grown rapidly in recent decades with a plethora of multispectral or hyperspectral (passive) or scanning (active) sensors [7–9]. Two important active sensor technologies include laser rangefinders and synthetic aperture radar (SAR) [10]. Systems with hyperspectral sensors have the ability to record electromagnetic energy in many different bands of the electromagnetic spectrum [11–14]. In addition, the high detail in the spectral information leads to improved accuracy [15]. Advances in the related, hyperspectral remote sensing, tools and the increase in the power of computing systems that has occurred in recent years have enabled the classification of large areas of land use/

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cover and discrimination with high accuracy [16]. Understanding landscape and natural resource changes is possible through the measurement and quantitative analysis of land use/land cover changes [17]. Various methods are used to classify them, such as traditional statistical classification methods, modern machine learning algorithms, deep learning techniques and geospatial techniques [18, 19].

To produce reliable LULC maps, various methodological approaches are used that involve the use of algorithms and models that analyze large datasets and predict changes in the Earth's surface [20–25]. Machine learning uses algorithms to analyze data, learn from that data and make formulated decisions based on what has been learned, such as support vector machine (SVM), random forest (RF) and K-nearest neighborhood (KNN) methods [25–31], achieving high accuracy [32]. Machine learning algorithms are much more reliable in area mapping than conventional algorithms due to their ability to solve challenging problems related to data processing [2, 33–39]. Also, the literature highlights in particular the high efficiency of SVM and RF in LULC mapping [32, 40]. For example, in a study conducted by Nhalia et al. [41] with data from the widely known hyperspectral images of Indian Pines, Pavia University and Salinas, both SVM and RF achieve very high accuracy rates. The same is highlighted also for example by Christovam et al. [42] where both classifiers were highly effective, with RF showing marginally higher accuracy than SVM.

The German Environmental Mapping and Analysis Program (EnMAP) imaging spectrometer mission was designed by a consortium of German institutions led by the German Aerospace Center (DLR) and launched on 1 April 2022. Its objective is hyperspectral remote sensing of the Earth,

specifically to measure, extract and analyse parameters related to the state and evolution of the Earth's surface and aquatic ecosystems on a global scale [43]. The objectives of the mission are to monitor changes in the Earth's environment, including land use, vegetation health, and water quality and soil conditions, to understand the impacts of climate change, to manage natural resources more efficiently by providing advanced hyperspectral data for the enhancement of Earth observation.

The present study aims to explore the efficiency of popular machine learning algorithms when combined with the advantages of hyperspectral EnMAP data in LULC mapping. More specifically, the objective is to evaluate the performance of SVM RF classifiers in LULC mapping using EnMAP hyperspectral data. For this purpose, as the study area is used the island of Milos in Greece as it denotes a typical Mediterranean environment with a landscape representing a variety of LULC categories. In addition, to our knowledge only one study has so far been published testing the ability of ENMAP data in with machine learning algorithms in LULC mapping in a Mediterranean [44]. As such, the present study adds information on the potential of hyperspectral EnMAP data in the field of LULC mapping.

2 Materials and methods

2.1 Study Area

The study area of this research study is the island of Milos (Fig. 1). Milos is a volcanic island in the Aegean Sea located in the southwestern part of the Cyclades and forms a dense

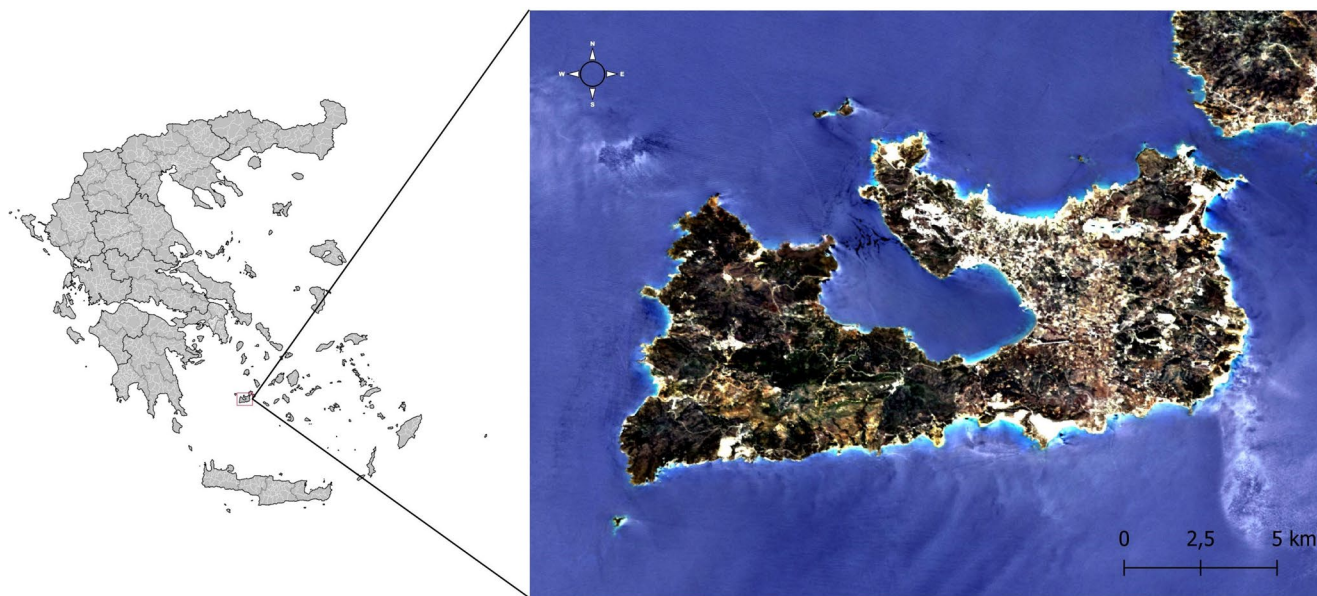


Fig. 1 Milos island, Greece (L2A EnMAP product, EOWEB geoportal <https://eoweb.dlr.de>)

cluster of islands together with Kimolos, Polyaiagos and Antimilos. It is the fifth largest island of the Cyclades with an area of 151 square kilometres and a coastline of about 125 km, with numerous bays and capes. Milos is characterized by a predominantly low and hilly morphology. The geography of the island varies greatly from one end to the other, influenced decisively by the particular geology of the area. The few lowlands are mainly found in the north-eastern part of the island, while in the south-western part the area is hillier due to the presence of large and numerous volcanic structures. The volcanic domes have shaped the topography, making this area more rugged and mountainous.

The typical Mediterranean environment of the island, reflected in the land use/cover categories, combined with the availability of EnMAP data for the area, played a role in its selection as a study area.

2.2 Data Used

The EnMAP operating with pushbroom technology was designed for measurements in the 420–2450 nm range via two separate spectrometers covering the VNIR and SWIR spectral regions. The sample is taken from 30 km x 30 km areas with a ground sampling distance (GSD) of 30 m. The average spectral sampling distance and resolution is at 6.5 nm in the VNIR and 10 nm in the SWIR. The ability to point outside the nadir up to 30° allows a return time of 4 days.

EnMAP data is available for free download from the EOWEB geoportal of the EnMAP ground segment. EOWEB is the interface to the German Satellite Data Archive. The EOWEB Geoportal is an Earth observation portal for multiple missions. Data can be requested via a user account. In this study, appropriate truncation was made in order to focus the research only on the land part of the island of Milos. The main characteristics of the dataset used are listed in the table below (Table 1).

The study used as auxiliary data from the Corine Land Cover (CLC) 2018 for the area of Milos. Their use also lies in the need for visual interpretation of the results. The

coordinate system used for the study is WGS84. CLC data were obtained in vector format through the Copernicus Land Monitoring Service (<https://land.copernicus.eu/en>) and used as guidance on land cover categories. In addition, Google Earth was used as a supplementary tool contributing to the photo interpretation of land use/land cover categories in the study area.

2.3 Methodology

Extracting information from satellite data is done by focusing on the spectral and spatial patterns that can be associated with land cover types. The contribution of classification algorithms in this process is that they exploit the spectral response values of each pixel, detected in the hyperspectral image under study, in order to classify it into its category based on the land surface characteristics [45].

To process the data and achieve the objectives of this study, we used the EnMAP Box-3 toolbox developed for processing and analyzing data recorded and processed by the EnMAP imaging spectrometer and available as an extension to the QGIS 3.34.7 software [46]. The methodology applied to classify land cover for the study area is shown below in diagrammatic form (Fig. 2).

2.3.1 Pre-processing

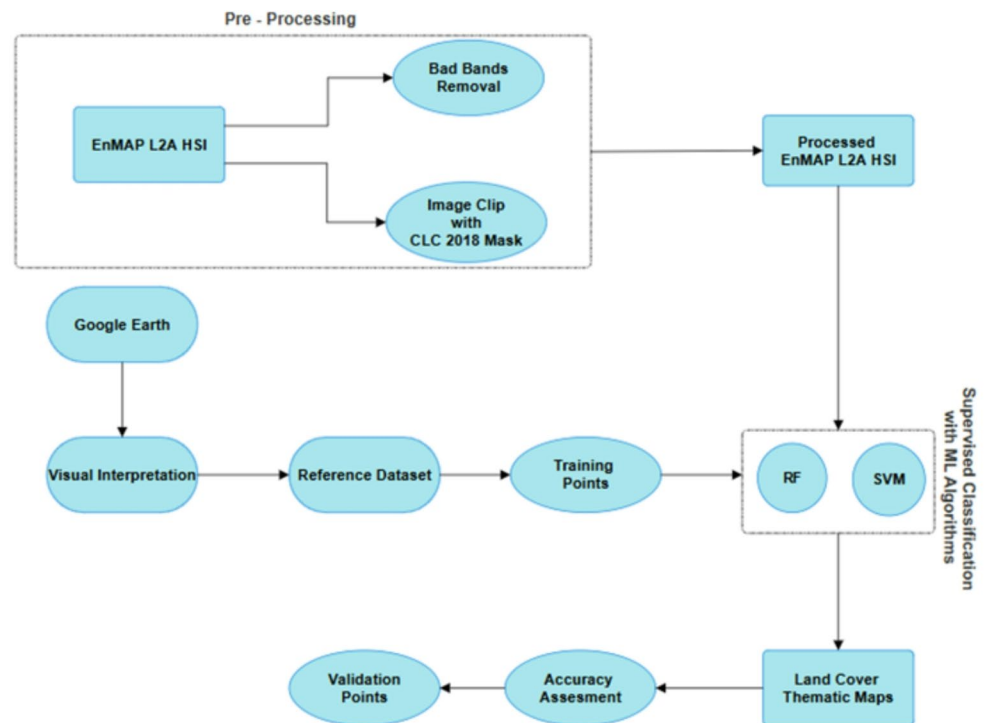
Hyperspectral data was acquired from the EOWEB Geoportal (EGP) (<https://eoweb.dlr.de/egp/>). The hyperspectral image processing level is L2A. This means that the product acquired for processing had already undergone atmospheric, radiometric and geometric corrections. The data were obtained in GeoTIFF format accompanied by metadata. The processing options were defined during the image selection process in the EOWEB Geoportal interface. A ground correction was selected to be applied and the type of atmospheric correction was defined as a combination of ground and water correction. This was done due to the presence of water bodies in different parts of the study area. The image resampling was performed using the Bilinear Interpolation method. In addition, the water type was selected, which was defined as clear and the reflection type, which was defined as normalized, and the cloud cover was also removed. The original L2A product consisted of 224 spectral channels with a range of 420–1000 nm (VNIR) to 900–2450 nm (SWIR). The accompanying metadata were utilized for its import into the EnMAP Box-3 toolbox.

When importing the image into the toolbox, an option to detect overlapping bands in the VNIR, SWIR regions and to remove the water absorption bands between 1358 and 1453 nm and between 1814 and 1961 nm was selected in order not to affect the analysis [47]. In addition, after

Table 1 Key characteristics of the enmap image acquired for the study area

Product	L2A
Landscape	Mediterranean environment
Atmospheric correction type	Combined (soil and water)
Geometric correction	Yes
Resampling	(Bilinear Interpolation)
Cloud Cover	Removal
Water type	Clear
Longitude/latitude	36.707504/24.42989
Projective system	WGS 1984 UTM 34 N – Greece
Download date	2023-06-03T09:51:54

Fig. 2 Workflow of the methodology adopted in the present study



reviewing the resulting hyperspectral image, a second “bad bands removal” was performed due to the existence of “noise” in the spectral response, mainly in the 900 to 1000 nm band, and finally resulted in the image on which the analysis and land use/land cover classification was performed, which consists of 170 spectral bands and is projected in the 34 UTM band of the WGS 1984 system. The last pre-processing step was the clipping of the sea surrounding the island for which no categorization was done as the study focuses on the island itself. This process was carried out in QGIS 3.34.7 software using the Corine Land Cover 2018 product as the water subtraction mask with appropriate conversion to the WGS 1984 coordinate system.

2.3.2 Support Vector Machines (SVM)

Support Vector Machines (SVM) were introduced by Cortes and Vapnik [48] in the early 1990s have shown promising results in applications at various fields including remote sensing and have been applied in many studies with hyperspectral data [25, 27, 42]. SVM is popular classification algorithm due to its generalization ability, good mathematical representations and geometric explanations and is therefore exploited for analyses involving classifications and regression. SVM algorithms are generally divided based on the types of kernel used to determine the hyperplane into linear (with a linear kernel) and nonlinear (with polynomial or radial basis functions) [49, 50]. A brief presentation of the equations of kernel-based SVMs is provided below [51]:

Linear kernel:

$$k(x_i, x_j) = i \cdot x_j \quad (1)$$

Polynomial kernel:

$$k(x_i, x_j) = (\gamma x_i \cdot x_j + r)^d, \gamma > 0 \quad (2)$$

Radial Basis function kernel:

$$k(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2) \quad (3)$$

Sigmoid kernel:

$$k(x_i, x_j) = \tanh(\gamma x_i \cdot x_j + r) \quad (4)$$

where γ, d, r are the parameters of SVM algorithms.

2.3.3 Random Forest

Random Forest (RF) is a popular ensemble learning method proposed by Breiman and used for classification and regression tasks [52]. It works by constructing multiple decision trees during training and extracting the class mode (classification) or mean prediction (regression) of the individual trees. A decision tree is a simple, rule-based model that divides data into subsets based on attribute values, with the goal of separating classes as discretely as possible. To classify a new feature vector, the vector is sorted by each of the

trees in the forest. Each tree gives a classification, i.e., it “votes” for a class, and the forest (the system) chooses the classification that has the most “votes” of all the trees. This algorithm shows many advantages such as high accuracy, efficient implementation on large datasets, and robustness to overfitting and easily storable structure for future use of pre-generated trees [52]. Among the learning methods, on which this algorithm works, the widely used methods are boosting and bagging [53]. The RF algorithm has been applied in several hyperspectral data classification studies [27, 42, 54].

2.3.4 EnMAP Hyperspectral Image Classification

To satisfy the study objectives, a supervised classification was chosen to map the land cover categories. This process was carried out using machine learning, SVM and RF algorithms. The selection of those classifiers was based on the review of the literature which highlights their excellent performance in the field of LULC classification [33, 55]. A radial kernel type SVM (RBF kernel type) was chosen to be applied as it balances data complexity and accuracy [56] and also because it requires the specification of a small number of parameters while giving accurate results [21–23, 57]. RF is often used for analyses that have hyperspectral images as input as it can provide satisfactory classification results without relying on an underlying probability distribution for the input data [58]. These two algorithms demonstrate excellent performance in LULC classification with a key advantage being their ability to handle complex datasets, such as those often found in EO datasets, either multispectral or hyperspectral.

2.3.5 Classification System Definition

Guided by Corine Land Cover (CLC) data from 2018, a land cover classification system was developed in seven distinct classes for the study area. This system resulted from the consolidation of CLC level 3 classes. Specifically, the subclasses ‘Olive Groves’ (221), ‘Non-irrigated Arable Land’ (211), ‘Complex Cultivation Patterns’ (242) and ‘Agricultural Areas’ (243) were entered uniformly as ‘Rural Areas’, as referred to in CLC level 1. The criterion for the merger was the similarity of the spectral signatures and, especially for the subcategory ‘Non-irrigated Arable Land’, the fact that it could not be identified as such since in order to verify that an area is not irrigated it is necessary to survey it in different periods of the same year, which is not the subject of the present study. In addition, the subcategories ‘Discontinuous urban fabric’ (112), ‘Port areas’ (123), ‘Airports’ (124) and ‘Sparsely vegetated areas’ (333) were registered under the class ‘Impervious surfaces’. The

criterion for consolidation was the similarity of the categories. This contributed to the increase in the area occupied by the resulting category. The sub-category “Mineral Extraction Sites” (131) was chosen not to be consolidated despite its similarity with the other impervious surfaces as it is a category of high importance for the island of Milos and therefore was decided to classify it as a separate category. Also, was added the category “Water Bodies” which is not included in the CLC categorization for Milos as we confirmed their constant existence in specific parts of the island through longitudinal observation of images of the area by the free software Google Earth. These adjustments were deemed necessary to facilitate the research due to the small area occupied by the island of Milos. The small area of the island implies a relatively small area coverage even for the dominant classes. Therefore, the adjustments made played an important role in screening the training and validation points without losing the representativeness of the main land cover classes of the area.

2.3.6 Training Points Collection

Google Earth very high resolution imagery available for the site and the different band combinations of the EnMAP hyperspectral image were used to select the appropriate points by photo-interpretation. This led to the definition of a small set of points. In particular, approximately 2000 points were selected as a set of training points through random sampling, aiming to be representative of each class. The same points were used to train each classifier. Also, the Jeffries – Matusita statistical measure [59] was applied through QGIS 3.34.7 software to test the discriminability of the different training sets selected representing the land cover classes. The calculated spectral discriminability values of the land cover categories indicate average discriminability with values centered around 1.4 on an overall scale of 0 to 2. The Jeffries – Matusita distance is a spectral distance metric that quantifies the difference or dissimilarity between two spectral signatures. This makes it particularly useful in improving the performance and accuracy of classification tasks.

2.3.7 Classification Application

In the context of this study, supervised classification was implemented using the widely used machine learning algorithms, SVM and RF, in order to produce thematic land cover maps of the study area with an approach similar to that of Lekka et al. [44]. The EnMAP Box-3 plugin’s EnMAP Box-3 workflow [46] and the functions of the scikit-learn library [60] embedded in this plugin. The performance of each model was evaluated by the accuracy process through

a set of validation points and its F1 score which is a derivative of the classification accuracy from the EnMAP Box-3 workflow.

The training and classification process for the RF and SVM algorithms were implemented in the EnMAP Box-3 toolbox via the “Random Forest Classifier” and “Support Vector Classifier (RBF kernel)” packages from the scikit-learn library. In order to complete the k-fold cross-validation process, the set of training points was split. The separation resulted in the training and test subsets, corresponding to 75% and 25% of the original training set of 2000 pixels, respectively. The automated validation process was followed through the GridsearchCV function, which is integrated in the EnMAP Box plugin, in order to select the values of the algorithm parameters. For the SVM algorithm, tests were performed within the default range of C and γ values [0.001, 0.01, 0.1, 1, 10, 100, 1000] and a CV value of the GridsearchCV function equal to 5. For the RF algorithm, the number of “decision trees” was set to the default value of 100. The following table shows the parameters used to classify the EnMAP image and the overall accuracy obtained (Table 2).

The RF algorithm parameters can affect its performance and refer to the quality of a split (criterion), the number of trees in the forest (n_estimators), the minimum number of samples required to split an internal node (min_samples_split), the minimum number of samples required to be at a leaf node (min_samples_leaf), the number of features to consider when looking for the best split (max_features) (If ‘sqrt’, then max_features is equal to the square root of the total number of features) and the maximum depth of the tree (max_depth) (if ‘none’, then nodes are expanded until all leaves are pure or until all leaves contain less than min_samples_split samples).

The SVM parameters allow to control the complexity and behavior of the classifier. Gamma (γ) defines how far the influence of a single training example reaches. Smaller values make the decision boundary smoother, which can reduce overfitting and the regularization parameter (C) which controls the strength of the regularization which is inversely proportional to C.

Table 2 Parameters of the algorithms selected for the classification process

Algorithm	Parameters	Value	OA
RF	Criterion	Gini	86%
	n_estimators	100	
	Max_depth	None	
	Max_features	Sqrt	
	Min_samples_split	2	
SVM (RBF)	Min_samples_leaf	1	90.60%
	C	1000	
	γ	0.001	

2.3.8 Classification Accuracy Estimation

The accuracy of the thematic land cover maps resulting from the classification process was assessed using various statistical parameters as described in the confusion matrix. The confusion or error matrix is a statistical analysis for classification accuracy, is considered representative of the entire study area and evaluates the performance of the classifier trained to classify the pixels of the hyperspectral image [61]. The statistical parameters that make up the error matrix are as follows: Overall Accuracy (OA), User’s Accuracy (UA), Producer’s Accuracy (PA), Kappa coefficient (Kc).

UA and PA are estimates of the actual, overall accuracy, which is for the whole study area [62, 63]. These accuracy (or error) values are related to the uncertainty inherent in the selection of validation samples [64]. OA estimates the overall classification accuracy and refers to the percentage of correctly classified validation pixels to the total number of validation pixels. PA indicates the percentage of correctly classified pixels within a given class and reflects the skip error, which represents pixels that are skipped from their reference class. UA represents the percentage of pixels classified in a particular class that actually belong to other classes [65]. The constant Kc measures the actual agreement between reference data and the classifier used to perform the classification, compared to the probability of agreement between the reference data and a random classifier. Compared to OA, Kc suggests a more conservative estimate than a simple percentage value [66]. The equations of the above parameters follow:

$$OA = \frac{1}{N} \sum_{i=1}^r n_{ii} \quad (5)$$

$$UA = \frac{n_{11}}{n_{irow}} \quad (6)$$

$$PA = \frac{n_{11}}{n_{icol}} \quad (7)$$

$$Kc = N \sum_{i=1}^r n_{ii} - \sum_{i=1}^r \frac{n_{icol} n_{irow}}{N^2} - \sum_{i=1}^r n_{icol} n_{irow} \quad (8)$$

where n_{ii} represents the number of pixels accurately classified into a particular category; the variable N refers to the total number of pixels present in the confusion matrix, while r denotes the number of rows, n_{icol} represents the sum of columns in the reference data, and n_{irow} denotes the sum of rows in the predicted categories.

To calculate the statistical parameters, validation points were selected by random sampling, which were created at the same time as the training points to ensure that they consisted of different pixels. They were approximately 700 in

number, following a ratio of 75% – 25% between training points and validation points respectively. The parameters OA, UA, PA were determined by the tool “estimating area and map accuracy for simple random sampling” embedded in the EnMAP Box-3 tool [67]. Then, these measurements were used to calculate the Kappa constant (Kc) using Microsoft Excel. Along with the accuracy estimate, an F1 score was generated representing how well the model classifies each class. The derived classification statistics provide valuable information as to the performance and reliability of the classification process and allow for a comprehensive evaluation of the accuracy of the results.

The RF and SVM algorithms were used to generate thematic land cover maps given the EnMAP hyperspectral image for the island of Milos. In order to evaluate their accuracy the same set of validation points was exploited. To evaluate the statistical significance of the results between the classifiers, the McNemar chi-square test (χ^2) [68] was used. This is a simple and accurate parametric test. The McNemar test uses two error tables to calculate the chi-square (χ^2) which compares the misclassifications of the two classifiers. In this way, cases are examined where one algorithm misclassifies a sample that the other algorithm classifies correctly and vice versa. This test is a simple approach for comparing different algorithms by class. The mathematical formula of the McNemar test follows:

$$x^2 = (f_{12} - f_{21})^2 / (f_{12} + f_{21}) \quad (9)$$

where f_{12} refers to the number of cases that are misclassified by one classifier but correctly by the other, while f_{21} refers to the reverse.

The McNemar test falls within the field of statistics. A statistically significant result is obtained when the value of the critical probability, p-value and χ^2 are within certain limits. In the case of the 2×2 matrix with the distribution of χ^2 having one degree of freedom, the value of χ^2 must be greater than 3.84. Similarly, p-value must take values less than the significance level. This level is set at 0.05 (5%). The test is based on whether the hypothesis that there is no significant

difference in the results between the two classifications (null hypothesis) is true or refuted [49]. That is, in the case of this study, both SVM and RF make the same wrong or right classification of the pixels of the hyperspectral image.

This statistical evaluation method has been widely used in the field of land use/land cover mapping [44, 69]. Adam et al. [70] used the test in the context of a land use/land cover classification study with RapidEye data in a coastal environment in order to compare the classifications of RF and SVM. However, the result showed that there is no significant difference in the results given by the algorithms.

3 Results

The two land use/land cover maps produced using ML algorithms are shown in Table 3; Figs. 3 and 4 summarizes the results of the classification evaluation in terms of its accuracy through the statistical parameters OA, Kc, PA and UA for both algorithms. The combination of the visual examination of the two thematic maps (shown in Figs. 3 and 4) with the CLC 2018 product (Fig. 5) and the statistical analysis shows that both classifiers performed well in accurately representing the spatial distribution of land use/cover types in the study area.

As can be observed from Table 3, the SVM classifier achieved UA values from 77% to 100% and PA values from 60% to 100% in 7 different land cover categories. The RF classifier showed a range of UA values from 72% to 100% and PA values from 61% to 100%. Overall, the SVM classifier showed an OA of 90.6% and a Kc value equal to 0.88 while the RF classifier achieved an OA of 86% and a Kc value of 0.82.

SVM showed very high accuracy in the categories of hardwood vegetation (UA 96%, PA 97.2%), salt marsh (UA and PA 100%) and water bodies (UA 100%, PA 92.3%) with F1 values respectively high in sclerophyllous vegetation (96.7%) and salines (100%). RF showed very high accuracy in the categories of natural grasslands (UA and PA 93.8%) and sclerophyllous vegetation (UA 90.4%, PA 90.8%), with water bodies achieving 100% UA and salt marshes 100% PA. The values of the above categories in the F1 score were

Table 3 The statistical results of the classification accuracy assessment

Land cover categories	SVM			RF		
	UA (%)	PA (%)	F1-score	UA (%)	PA (%)	F1-score
Impassable Surfaces	88	60	0.882	72	78	0.750
Mineral Mining Areas	77	89	0.770	81	61	0.696
Agricultural Areas	89	96	0.924	84	87	0.857
Natural Grasslands	89	96	0.921	94	94	0.938
Sclerophyllous Vegetation	96	97	0.977	90	91	0.906
Salines	100	100	1	87	100	0.93
Water bodies	100	92	0.96	1	85	0.917
Overall Accuracy (%)			90.6			86
Kappa coefficient			0.879			0.818

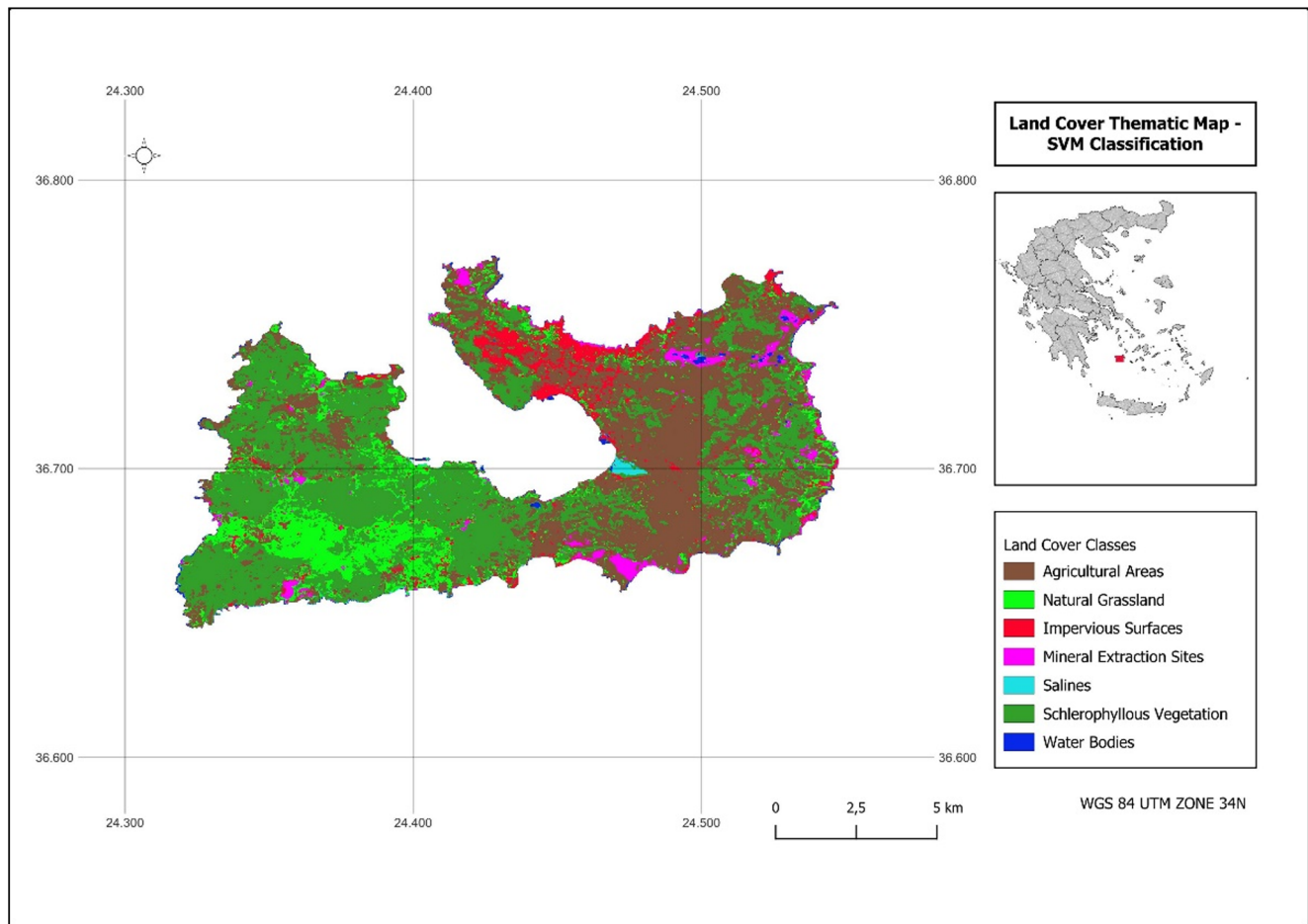


Fig. 3 Thematic land use/land cover map as derived from the supervised classification of the SVM algorithm based on the EnMAP satellite hyperspectral image acquired by the EOWEB Geoportal (<https://eoweb.dlr.de>) – Google satellite background

93.8% for natural pastures, 93% for salt marshes, 90.6% for sclerophyllous vegetation and 91.7% for water bodies.

The categories with the lowest values in the SVM classification were impervious surfaces (UA 88.2%, PA 60%) and mineral extraction sites (UA 77%, PA 89.1%) as in the RF classification with values of UA 72.2%, PA 78% and UA 81.3%, PA 60.9% respectively. The F1 score was lowest in both classifications in the category of mineral extraction sites with a score of 88.2% in the SVM classification and 75% in the RF classification.

The statistics for the other categories are as follows: for the SVM classifier the agricultural sites category achieved UA 89.3% and PA 95.8%, natural pastures UA 88.6% and PA 95.9% while for the RF classifier the values in the agricultural areas category were UA 84.2% and PA 87.2%.

Below is a figure showing the land use/land cover classes as derived from the merger of the CLC 2018 classes when determining the classification system (Fig. 5).

In order to perform a more extensive comparative evaluation of the classifier performance, the McNemar's test was

also applied based on a 2×2 matrix generated to calculate the number of pixels correctly classified by SVM and incorrectly classified by RF and vice versa. The test resulted in the p-value which was 0.01 and the χ^2 value which was 6.6.

All in all, the results of this study highlight that both classification approaches considered have very promising prospects for achieving high classification accuracy in discriminating land cover classes when combined with the EnMAP hyperspectral image. Furthermore, the advantages of the high spectral resolution of the EnMAP sensor are also indicated by the values of the statistical parameters of overall accuracy (OA) and Cohen's constant (Kc).

4 Discussion

Machine learning algorithms are widely used in LULC mapping. Many machine learning-based techniques are used in combination with hyperspectral data in order to highlight and exploit the potential of these data [71]. As

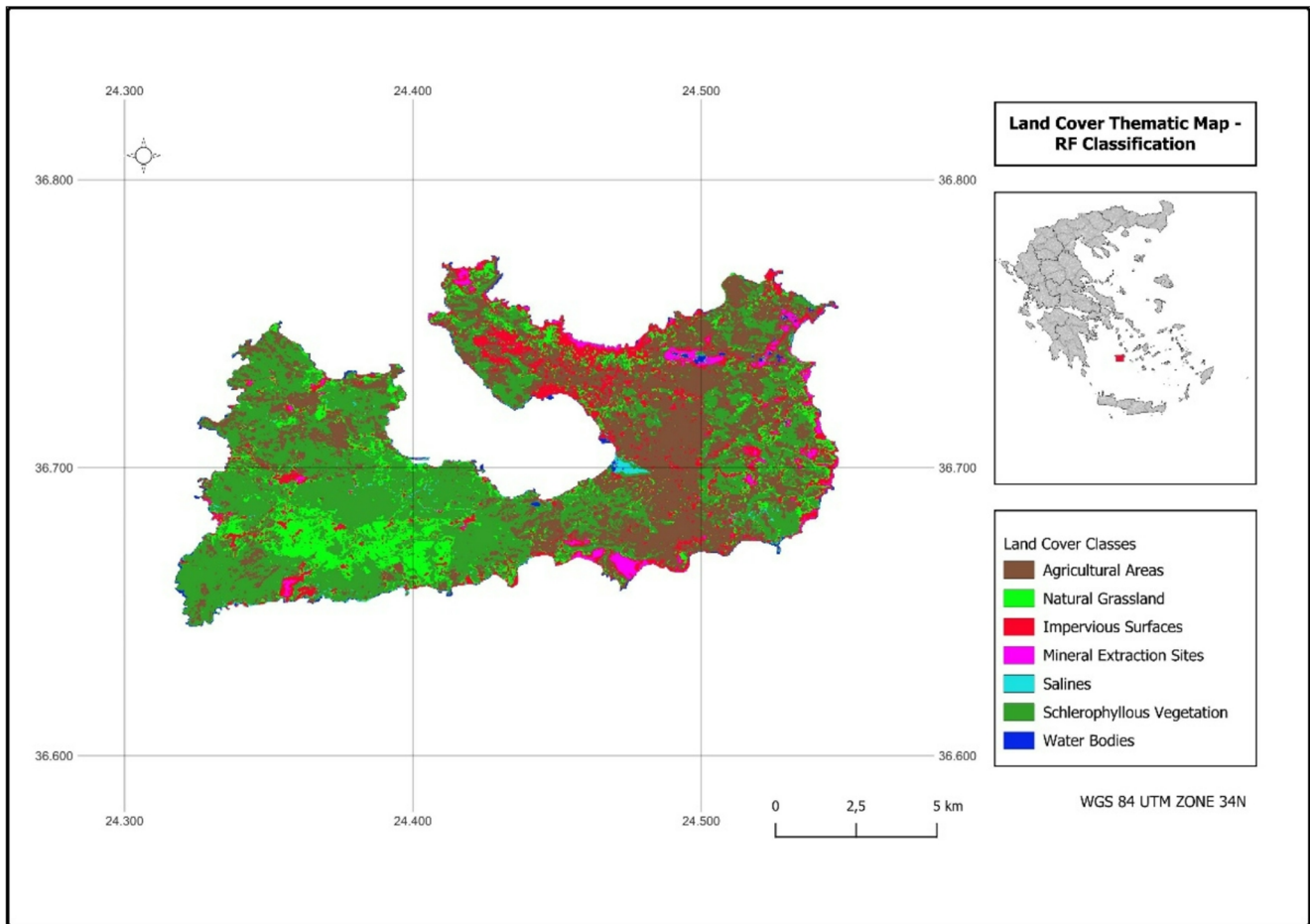


Fig. 4 Thematic land use/land cover map as derived from the supervised classification of the RF algorithm based on the EnMAP satellite hyperspectral image acquired by the EOWEB Geoportal (<https://eoweb.dlr.de>) – Google satellite background

a consequence, findings related to the evaluation of the performance of these techniques in LULC mapping have emerged and have been documented in several studies [25, 54] which were conducted at different scales and with different settings. According to Aryal et al. [72], the integration of RF and SVM, among other machine learning algorithms, in hyperspectral data processing is becoming increasingly common, covering a wide range of applications with excellent performance in LULC mapping.

This study evaluates the effectiveness of EnMAP hyperspectral data combined with the popular RF and SVM algorithms with RBF kernel type algorithms for land use/cover mapping in a typical Mediterranean environment. The results of both classifiers were accurate for defining seven land use/land cover classes involving impervious surfaces, agricultural areas, water bodies, salines, sclerophyllous vegetation, natural grassland. The dominant classes in the study area in terms of extent are those of hardwood vegetation and agricultural areas, as reflected in both thematic maps and also confirmed by the CLC 2018 product.

The SVM classifier achieved higher accuracy compared to the RF classifier, in this study, as indicated by the statistical parameters Overall Accuracy (OA) and Kappa coefficient (Kc). The McNemar's test helped to further evaluate the accuracy of the performance of the two algorithms. The test revealed that the result of the two classifications is statistically significant, judged by the p-value (<0.01) which was less than 0.05. Similarly, the value of χ^2 was greater than the critical value of 3.84. Therefore, the null hypothesis, which claims that there is no significant difference in the classification results performed by the two algorithms, is refuted by highlighting that the two algorithms have different performance when trained with the same set of samples. Therefore, the results of both algorithms are statistically significant.

In the case of SVM, the best performance was achieved in the categories of sclerophyllous vegetation (UA 96%, PA 97.2%), salines (UA and PA 100%) and water bodies (UA 100%, PA 92.3%). The overall accuracy was 90.6% and the Kc value was equal to 0.88. The RF classifier

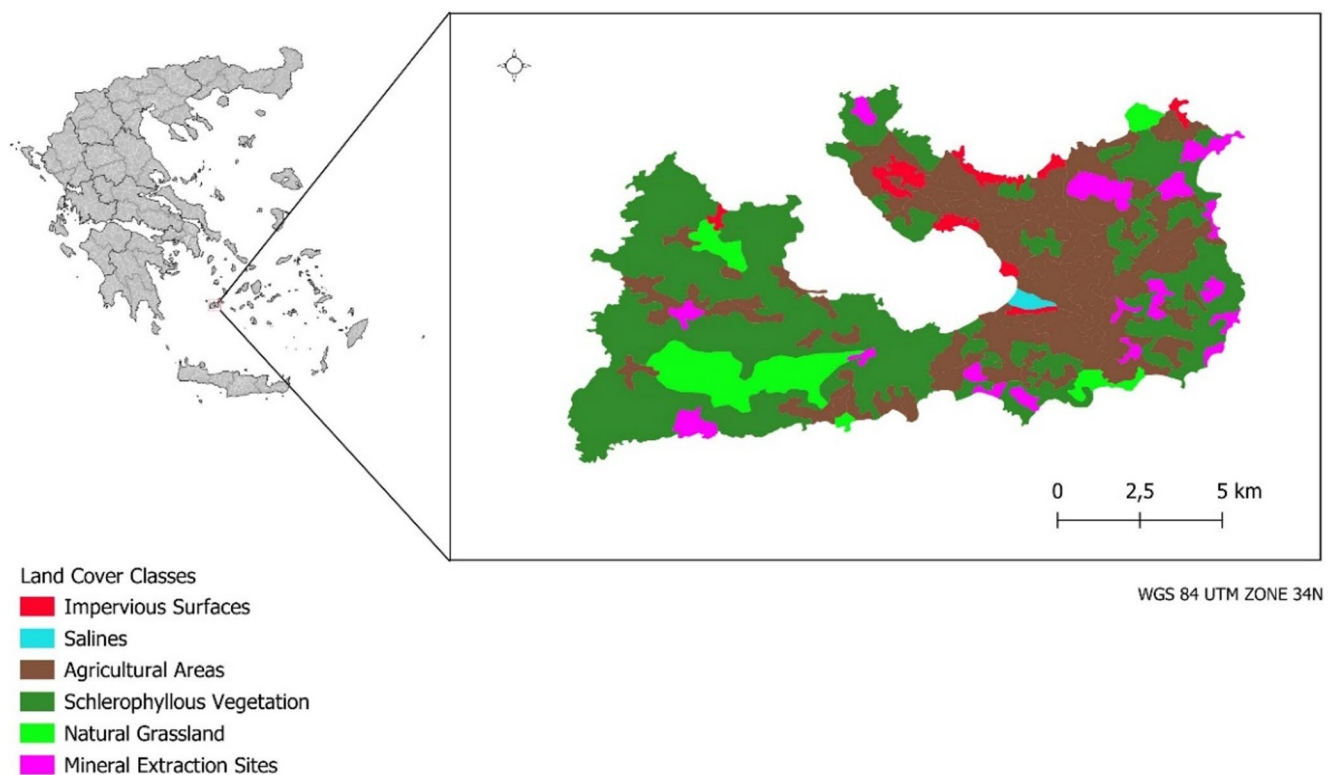


Fig. 5 Land use/cover categories of the island of Milos as derived from the consolidation of the CLC 2018 classes

performed better in the categories of natural grassland (UA and PA 93.8%) and sclerophyllous vegetation (UA 90.4%, PA 90.8%), with water bodies achieving 100% UA and salt marsh 100% PA, while the OA was 86% and Kc value equal to 0.82. The above percentages highlight the agreement between the predicted land cover classes and the reference classes. In both classifiers, the water bodies and salines and salt marsh classes were almost perfectly identified, which is related to the small number of samples that could be obtained for these classes, their homogeneity, combined with their very small area. The lowest values for both classifiers occurred on impervious surfaces and mineral extraction sites which is related to their similarity in spectral response. This should take into account the fact that the 30 m spatial resolution of EnMAP images makes it difficult to distinguish between areas with similar features. Agricultural areas showed very good values for both classifiers.

This study consists to the authors knowledge one of the first attempts to evaluate the potential of the state-of-the-art EnMAP hyperspectral data for land use/cover mapping with the contribution of machine learning algorithms. In addition, this effort is one of the first efforts in a Mediterranean context. A related study with hyperspectral EnMAP data has been done by Lekka et al. [44]. Based on this, the comparison of the results of this study can be made with the results of this study and those obtained from studies with

hyperspectral data from other sensors [73]. For example, Delogu et al. [74], evaluated the ability to classify permanent agricultural crops in a highly fragmented agricultural area in Campania, Italy, using PRISMA data, the EnMAP Box plugin and with an RF classifier, among others, achieving an overall accuracy of 88% and Kc of 87%. Similarly, Lamine et al. [75], used Hyperion hyperspectral data combined with the SVM algorithm to evaluate spatio-temporal land use/land cover changes in central Greece. The study achieved a maximum OA of 92% and a maximum Kc of 91%. The results of other studies also highlight the higher accuracy achieved by SVM algorithms compared to RF [11, 41] although the performance of both is quite satisfactory. A typical example is the research by Alcolea et al. [76] based on state-of-the-art machine learning algorithms for supervised land use/cover classification, including SVM and RF. Different land cover classes were considered for different hyperspectral datasets, namely Indian Pines, Kennedy Space Center and Salinas Valley from AVIRIS, University of Pavia from ROSIS and University of Houston from CASI. The results confirm that the RF algorithm achieves lower accuracy values while SVM (RBF kernel) outperforms. The values for RF and SVM range from 73 to 90% OA and 70–89% Kc and from 77 to 94% OA and 92% Kc respectively. Finally, Lekka et al. [44] proceeded to classify LULC using the RF and SVM algorithms with hyperspectral

EnMAP data in a typical Mediterranean environment. The results showed overall high classification accuracy by both classifiers, however, with SVM leading the way. In particular, SVM demonstrated an OA of 90.5% and Kc of 0.89 and RF OA of 87.5% and Kc of 0.86. The findings of this study are in agreement with the findings of the present study. The importance of the agreement between these two studies is particularly significant as both used hyperspectral EnMAP data and the study area was in the category of typical Mediterranean environment.

An important element of the surveys is the correct choice of algorithm related to the type of classification sought and also to the input data and the set of classes. Also, the tuning of the parameters of each algorithm can affect the performance of the classification. However, based on literature, RF can also perform satisfactorily with the default parameters [77]. In this research, the default parameters obtained from the automated k-fold cross-validation through GridsearchCV using the default value range were selected. In terms of computational power, the algorithms did not show large differences in terms of classification execution time. The analyses were performed in QGIS 3.34.7 software through the EnMap - Box 3 tool. The RF algorithm was shown to be faster, with a difference of a few minutes, as expected based on relevant reports in the literature [33, 78]. The execution time of both classifications was not particularly long (around 20 min), which is probably related to both the small number of training points and the parameterization chosen.

The classification of hyperspectral remote sensing images has made significant progress in recent years, highlighting more and more possibilities. However, along with the potentials come challenges associated with this type of data. The main challenge remains the complexity and volume of hyperspectral data leading to high operational and computational costs. This characteristic is linked to the process of data processing, storage and analysis. The high dimensionality of hyperspectral images, with hundreds, narrow, continuous, spectral bands make their processing difficult and time-consuming, requiring the corresponding software [16]. Therefore, challenges are presented at the data preparation stage. Such as, the lack of labeled training samples which makes it very important, difficult and time consuming for the user to collect them in order to make use of the appropriate features for each land cover category.

In the context of this study, an important limitation was identified, which together with the complexity of the data, already mentioned, makes the analysis task more difficult. This is linked to the small set of training samples due to the small size of the study area and the high level of landscape fragmentation. This limitation leads the models to overfitting to the training points, making it difficult to generalize

and therefore use the model in other areas [79]. However, from the outset, it is judged that the choice of RF and SVM algorithms was appropriate as they demonstrate good performance on small sample sets [42]. Furthermore, the spatial resolution of EnMAP poses challenges regarding the discrimination between classes with similar spectral features and also spectral heterogeneity within the same class [49]. The fact that both classifiers used do not operate at the sub-pixel level highlights the mixing problems and increases the probability of misclassification at pixel level [21–23].

5 Concluding Remarks

The aim of the study was to evaluate the effectiveness of combining EnMAP hyperspectral imagery with the popular machine learning algorithms, RF and SVM to produce thematic land use/land cover maps in a typical Mediterranean environment.

The results of the analysis are promising for the role of EnMAP hyperspectral data in the field of land use/cover mapping. The main conclusions drawn from this study include:

- The comparison of the two classifiers highlights superiority of SVM over RF as the OA and Kc values were higher. SVM showed OA 91% and Kc 88% against OA 86% and Kc 82% shown by RF respectively. The superiority of SVM was also documented by the McNemar test. Nevertheless, the RF classifier remains very accurate, based on the overall statistical parameters reported and the values achieved for individual classes as presented in the corresponding table. Therefore, the suitability of both algorithms in LULC mapping using hyperspectral EnMAP data is confirmed.
- The visual comparison of the thematic maps resulting from the two classifications with the Corine Land Cover product highlights the general agreement in terms of the spatial distribution of categories. This comparison highlights the relativity resulting from the different characteristics of the products which makes a direct comparison impossible. However, studies investigating the effectiveness of hyperspectral data in LULC mapping can enhance the effort to improve and rapidly update operational LULC products that play an important role in key areas of modern life.
- In the classification process 170 spectral channels of EnMAP image were exploited highlighting the large volume of spectral data leading to time-consuming processing and analysis on the benefit of improving the targets separability on the ground. The results obtained must take into account the small number of training and

validation samples combined with the fact that no techniques were carried out to reduce the data volume and increase the number of samples. However, the innovation of the large number of spectral bands in hyperspectral data lies in its ability to provide a far more detailed and nuanced view of the Earth's surface compared to traditional multispectral data and this vast amount of spectral information captured by hyperspectral sensors allows for the extraction of detailed features that can enhance classification accuracy.

- The open-source software QGIS 3.34.7 with the use of EnMAP - Box allowed satisfying the study objectives and performing the analysis. The combination of open-source software with the availability of open access hyperspectral data and machine learning algorithms proves a valuable tool in remote sensing related topics research.

This study contributes to the existing knowledge of hyperspectral telescopic data processing by adding information on the performance of recently available EnMAP data in its use together with established machine learning algorithms. An important aspect of the study is that it adds land use/land cover data in a typical Mediterranean environment that can be compared in future research and the results can be compared with similar results using EnMAP data in other environments in order to draw conclusions about their reliability. This remains to be seen.

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Data Availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Competing Interests The authors declare no competing interests.

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